

Gauge-Coupling Parameters and Forced Uniqueness under Admissible Redescription

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Abstract

This paper examines whether gauge-coupling parameters can carry forced uniqueness under admissible redescription. We show that they cannot: raw numeric gauge-coupling values are chart-level quantities whose values vary under admissible reparameterization, and therefore cannot serve as standing-bearing invariants. The result is not merely a restatement of scheme dependence. It follows from minimal evaluability constraints: same-object comparison requires identity stability, cross-presentation comparability, and redescription-invariant standing. We then prove that every same-scope route to numeric uniqueness reduces to invariant fixation, invariant exclusion, representative choice, witness or completion import, or scope change. These routes either fix tensor-invariant content, alter the target problem, or collapse to closed selector/import mechanisms. Numeric coupling collapse is therefore not an admissible uniqueness target.

The positive result is that uniqueness reappears at the correct role level. For genuinely co-scoped admissible regimes claiming authority over the same object-class, overlap is necessary; divergence on overlap is inadmissible; and the admissible representative is forced to coincide up to fixed normalization. Thus the paper replaces numeric coupling uniqueness with overlap-level forced uniqueness: gauge-coupling parameters remain skin-level coordinates, while invariant tensor content and shared-domain representatives carry admissible physical standing.

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Reader orientation. The paper has one negative result and one positive result. The negative result is that a raw chart-numeric gauge coupling is not forced to a unique value by admissibility. The positive result is that, when two admissible regimes genuinely claim the same object-class on a common domain, their admissible physical content cannot diverge on the overlap. Thus the uniqueness proved here is overlap-level forced coincidence, not numeric coupling determination.

1 Global Constraints

All reasoning proceeds under the following fixed admissibility constraints:

- (G1) **Standing conservation:** standing-bearing identity may not change under admissible transformation.
- (G2) **Redescription stability:** admissible redescriptions preserve standing identity and tensor compatibility.
- (G3) **Closure:** admissible composition admits no divergence on shared (overlap) domains once normalization is fixed.
- (G4) **Bivalence / fail-closed:** admissibility is binary; violations are terminal.
- (G5) **No new primitives:** no new standing-bearing structure may be introduced ad hoc.

In addition, the following operator families are treated as *CLOSED as certifiers of standing* (KFRT: these closures do not refute the corresponding mathematical formalisms; they block only operator-level attempts to *generate* standing by those means):

- Meta-foundational re-anchoring (operator-level grounding by external authority)
- Model-theoretic existence certification (existence-by-model as standing certifier)
- Compactness and saturation as operators (completion/existence certification)
- Typicality and measure as operators (selection/legitimation by weighting)
- Choice/selection operators (nontrivial selection on undistinguished inputs)
- Genericity/forcing operators (generic output as operator power)
- Operator totalization (canonical completion that generates standing)

These closures are used as fixed upstream results; they are restated in Section 4 in minimal theorem schema form.

2 Framework Dependencies and Scope Lock

This manuscript uses the AASC framework in one narrow way: it relies on already-fixed closure results that constrain what may count as a same-scope admissible maneuver. The present paper does not re-prove the full framework and does not introduce a new foundation. It states the required supports explicitly so that later appeals are dependency-transparent rather than hidden stipulations.

2.1 Fixed gauge-sector comparison class

Definition 2.1 (Fixed gauge-sector comparison class). *A challenge, redescription, or rival construction remains in the same gauge-sector target only if it preserves all of the following components simultaneously:*

- (C1) *the admissibility anchors in Section 1, including fail-closure, standing conservation, and no new primitives;*
- (C2) *the object-class under adjudication: raw chart-numeric gauge-coupling labels versus tensor-level gauge-sector compatibility data are not interchangeable targets;*
- (C3) *the standing identity quotient induced by admissible redescription (Definition 3.2);*
- (C4) *the tensor package being held fixed, such as representation compatibility, anomaly/consistency constraints, locality/observable-compatibility constraints, and any explicitly fixed normalization convention;*
- (C5) *the licensed continuation and translation structure, including RG evolution, admissible scheme/chart change, and overlap translations when the same object-class is asserted; and*
- (C6) *the carrier on which anchor/tensor/skin role assignment is evaluated.*

Changing any one of (C1)–(C6) is a scope change. It may define a different problem, a different theory family, or a different tensor package, but it is not a same-scope counterexample to this paper’s closure claims.

Lemma 2.2 (Scope-shift isolation). *Any purported refutation of this paper that changes one of the components in Definition 2.1 either exits the target comparison class or introduces illicit scope transport if it continues to present itself as same-scope.*

Proof. The fixed target is individuated by the six preserved components. If a maneuver changes the anchors, object-class, standing quotient, tensor package, continuation/translation structure, or role carrier, then it has changed the object under audit. Treating the new object as though it were still the old one requires transporting standing from one scope to another without an admissible witness. That violates standing conservation and no-new-primitives. Hence such a maneuver is not a same-scope refutation. $\square$

2.2 Imported source normal form

The following dependency map records the AASC results used in this paper. These references function as internal framework dependencies, not as appeals to external authority; their local role is to identify which upstream closure result is being invoked.

Framework result	Local use in this paper
AMetric comparison-class lock and no-selector/no-parameter boundary	Blocks continuous, discrete, metric, typicality, canonical-representative, and gate-chooser attempts from carrying boundary authority inside the fixed comparison class.
Anchor/Tensor/Skin role factorization and skin non-authority	Forces raw chart labels, notation, parameter coordinates, and presentation choices to remain skin unless they change standing-invariant tensor compatibility.
Standing-Admissibility identity closure and no-second-classifier closure	Blocks dual same-domain gates and any auxiliary classifier that tries to decide which admissible representative governs on the same overlap.
Minimal Conditions for Admissible Construction	Supplies the minimal assessability boundary: determinate reference, standing, and non-repairable invalidity are forced by identity-preserving construction rather than added as protective clauses.
Closure by Exhaustion for same-scope operators	Supplies the audited-kernel theorem: every non-bookkeeping same-scope operator depends on a closed kernel family, and interaction among closed families creates no new admissible family.
Gauge/Constraint/Quantum structural-origin bridge	Fixes the physics-facing posture: constrained/gauge-type structures are treated at the quotient/tensor level, while formal presentations and chart choices are lawful realizations when they preserve the same theorem object.
Fixed-domain/bivalence/fail-closed continuation layer	Supplies the binary, no-repair, no-deferred-admissibility, and no-silent-redescription background used whenever violations are treated as terminal.

Lemma 2.3 (Framework normal form for the present target). *Within the fixed gauge-sector comparison class, every same-scope admissibility-bearing difference falls into exactly one of the following dispositions: anchor change, tensor change, or skin/bookkeeping. Any purported standing-generative operator that is not already fixed by anchor/tensor invariants is inadmissible.*

Proof. The role-factorization result exhausts descriptive work into anchors, tensors, and skin on a fixed carrier. The AMetric boundary blocks selector, metric, weighting, index, and canonical-representative authority at the boundary. The standing/admissibility identity closure blocks a second same-domain gate. The operator-exhaustion result closes direct operator families and interaction-generated variants. Therefore a same-scope difference either changes preconditions (anchor), changes invariant compatibility/standing content (tensor), or changes no admissibility-bearing outcome (skin/bookkeeping). A non-bookkeeping operator not fixed by those invariants must import a selector, witness, boundary, weighting, authority, completion, or other audited kernel, all of which are closed as standing certifiers. $\square$

Corollary 2.4 (No numeric selector route). *A rule that selects a unique raw numeric gauge-coupling value from a non-singleton admissible chart domain is either bookkeeping because the value was already fixed by tensor invariants and normalization, or inadmissible because it imports a selector, metric, weighting, canonical representative, empirical datum, or scope-changing boundary.*

Proof. A raw numeric value is a representative inside a chart. If it is already fixed by standing-invariant tensor data, selection adds nothing. If it is not so fixed, any rule distinguishing it from other admissible representatives is a same-scope selector or a scope-changing import, both closed by Lemma 2.3. $\square$

3 Core Definitions and Role Discipline

3.1 Role glossary (anchor/tensor/skin in operational terms)

To prevent category ambiguity, we record the intended role discipline in operational (non-metaphorical) terms:

- **Anchors:** non-negotiable admissibility constraints (e.g., bivalence/fail-closed, no illicit scope transport, no new primitives, standing conservation).
- **Tensors:** load-bearing compatibility relations fixed by anchors (e.g., consistency conditions such as anomaly cancellation, representation compatibility, locality constraints as role constraints, and normalization-fixing conventions when they are explicitly stipulated).
- **Skin:** expressive/chart layer (notation, coordinates, parameterizations). Example: a coupling reparameterization $g \mapsto \tilde{g} = f(g)$ with f invertible on the admissible domain is skin-level if tensor-compatibility relations are unchanged.

The central nontrivial stance of the paper is not that “parameters are meaningless,” nor that observables cannot depend on RG-invariant trajectory data. The stance is narrower and stronger: *raw numeric labels that vary under admissible reparameterization cannot be promoted to standing-bearing tensor identity without importing illicit structure* (Definitions 3.1–3.8 and Corollary 2.4).

3.2 Formal core definitions

Definition 3.1 (Admissible redescription). *An admissible redescription is a representational change satisfying:*

- (R1) **Role preservation:** *anchor roles remain fixed; tensor roles remain fixed.*
- (R2) **Compatibility preservation:** *tensor-level compatibility relations are unchanged.*
- (R3) **Skin confinement:** *all differences are confined to skin-level expression (notation, coordinates, charting).*

Redescription is representational only; it does not generate standing and cannot repair standing failure.

Definition 3.2 (Standing identity). *A standing identity class is an equivalence class of representations under admissible redescription (Definition 3.1) with fixed admissible invariants. Two representations denote the same standing identity iff they lie in the same equivalence class.*

Definition 3.3 (Standing-bearing quantity). *A quantity q is standing-bearing iff:*

- (S1) *q is invariant under admissible redescription, and*
- (S2) *variation of q changes tensor compatibility or standing identity.*

If q varies under admissible redescription while compatibility is preserved, q is not standing-bearing.

Definition 3.4 (Discriminative same-object evaluation). *A regime performs discriminative same-object evaluation when it issues determinate admissibility, identity, compatibility, or physical-content verdicts about presentations that may represent the same standing object-class. Such evaluation is nondegenerate only when some verdicts can differ when standing-relevant tensor data differ, while identity-preserving presentation changes do not by themselves change the verdict.*

Theorem 3.5 (Minimal evaluability necessity). *Any regime capable of discriminative same-object evaluation must supply identity stability, cross-presentation comparability, and nondegenerate verdicts. Consequently, standing conservation, redescription stability, bivalence/fail-closure, and no-new-primitives are minimal evaluability conditions for same-object claims, not optional framework preferences.*

Proof. A discriminative same-object verdict has three irreducible requirements. First, the object of evaluation must remain stable across the presentations being compared; otherwise the verdict equivocates between different targets and no same-object claim is made. Second, the presentations must be comparable under a fixed translation or redescription discipline; otherwise they are isolated descriptions and no cross-presentation identity or compatibility verdict is possible. Third, the verdict must be nondegenerate: if every outcome is permitted, evaluation has no force; if an outcome is privileged without invariant grounds, a selector has been added; if a failed outcome is repaired after the fact, failure is not determinate. These alternatives force standing conservation, redescription stability, binary fail-closure, and no-new-primitives as the minimal conditions under which same-object evaluation can be meaningful. A regime may decline to make same-object claims, but once it makes them it cannot discard those conditions without losing evaluability. $\square$

Theorem 3.6 (Redescription-invariance necessity). *Let R be a nondegenerate fixed-domain regime in which two presentations may be compared as representations of the same standing identity. If a quantity q is used to carry standing-bearing identity or tensor-compatibility force on that domain, then q must be invariant under admissible redescription. If q changes under admissible redescription while tensor compatibility is preserved, then treating q as standing-bearing is inadmissible.*

Proof. Let x and y be admissible redescriptions of the same standing identity. Suppose $q(x) \neq q(y)$ while the anchor roles and tensor-compatibility relations are preserved. If both values are allowed to carry standing-bearing force, then the same standing identity receives divergent same-domain identity or compatibility verdicts, violating standing conservation and bivalence. If only one value is privileged, a higher-order selector is required to decide which presentation governs; that is exactly the no-second-classifier/selection route closed by Section 2 and Theorem 4.5. If the regime instead declares x and y to be different standing identities solely because q differs, it changes the standing quotient rather than preserving the original comparison class. Thus any quantity with standing-bearing force must be redescription-invariant. Conversely, a redescription-varying quantity whose variation leaves tensor compatibility fixed contributes only presentation data and cannot carry standing. $\square$

Corollary 3.7 (Standing-bearing criterion is not protective). *Definition 3.3 is not an ad hoc filter against gauge couplings. It records the minimal evaluability condition for any same-domain quantity claimed to carry standing: invariance under identity-preserving presentation change and tensor-level effect when varied.*

Proof. Theorem 3.6 forces the invariance clause. If a quantity is invariant but its variation changes neither standing identity nor tensor compatibility, then it performs no admissibility-bearing work on the fixed domain and is bookkeeping by Definition 3.10. Hence both clauses of Definition 3.3 follow from the possibility of identity-preserving evaluation rather than from a protective definition. $\square$

Definition 3.8 (Chart-numeric coupling value). *A chart-numeric coupling value is the real-number label assigned to a gauge-coupling coordinate after a choice of scheme, chart, reference scale, and normalization convention. It is not, merely as a number, a standing-bearing quantity. It becomes standing-bearing only through redescription-invariant content satisfying Definition 3.3, such as tensor-level compatibility effects or properly invariant observable data.*

Theorem 3.9 (Admissible physical content is invariant content). *Within the fixed gauge-sector comparison class, a same-object claim has admissible physical content only through redescription-invariant tensor or observable data. Hence an overlap-level forced-coincidence result is not merely notational: it is uniqueness of the admissible physical content in that comparison class. Any alleged physical difference between admissibly redescribed representatives is either an invariant tensor difference, in which case the same-object hypothesis has failed, or a skin difference, in which case it has no physical standing.*

Proof. Suppose two representatives are related by admissible redescription. By Definition 3.1, anchor roles and tensor compatibility are preserved and all differences are confined to skin. If a purported physical claim distinguishes the two representatives despite that preservation, the distinction must either depend on a redescription-variant label or on added invariant structure. In the first case, Theorem 3.6 blocks the distinction as standing-bearing content. In the second case, the added invariant changes the tensor package or standing quotient and therefore exits the same-object comparison. Thus admissible physical content is exactly the invariant content available under the fixed comparison class. Forced coincidence of the admissible representative on overlap therefore coincides with physical-content uniqueness at that admissibility level, without making an extra ontic or explanatory claim. $\square$

Definition 3.10 (Bookkeeping equivalence). *A transformation/operator is bookkeeping-equivalent iff it introduces no new standing-bearing invariants and adds no admissible operator power beyond re-expression of already fixed structure.*

4 Upstream Closure Results (Operator Families)

This section records the minimal closure schemas used later. They are taken from the framework dependencies summarized in Section 2 and are used only in their operator-relative sense: they block attempts to *certify standing* by the named method; they do not refute the corresponding methods as formal mathematics.

Theorem 4.1 (Meta-foundational re-anchoring closure schema). *Any operator whose action purports to generate standing by appeal to a meta-foundational authority imports standing not generated by the operator’s own standing-preserving action. Hence no such operator is admissible as a standing certifier. In short: “authority” cannot be a generator of standing; it can only be imported.*

Theorem 4.2 (Model-theoretic existence closure schema). *Any operator that asserts existence or legitimacy by appeal to the existence of a model/structure imports a witness whose standing is not generated by the operator. Hence no such operator is admissible as a standing certifier. In short: “a model exists” cannot certify standing without importing a standing-bearing witness.*

Theorem 4.3 (Compactness/saturation closure schema). *Any operator that asserts existence/realizability/completeness by appeal to compactness or saturation imports standing not generated by admissible action. Hence no such operator is admissible as a standing certifier. In short: compactness/saturation may prove formal existence, but cannot certify standing without imported witnesses.*

Theorem 4.4 (Typicality/measure closure schema). *Any operator that asserts legitimacy/existence/selection by appeal to typicality or measure imports standing not generated by admissible action. Hence no such operator is admissible as a standing certifier. In short: weighting/typicality selects by an external measure and therefore cannot generate standing.*

Theorem 4.5 (Choice/selection closure schema). *Let D be an admissible domain with nontrivial admissible multiplicity (undistinguished inputs). Any operator purporting to select a distinguished element is either inadmissible or bookkeeping-equivalent (Definition 3.10). Hence no nontrivial selection operator is admissible. In short: any genuine selector is either redundant (already fixed) or illicit (oracle/distinguishing indistinguishables).*

Theorem 4.6 (Genericity/forcing closure schema). *Any operator purporting to output a genuinely generic object on an undistinguished nondegenerate forcing context is either inadmissible or bookkeeping-equivalent. Hence genericity/forcing cannot generate admissible operator power as a standing certifier. In short: genericity cannot create admissible standing-bearing uniqueness; it either imports a context or collapses to bookkeeping.*

Theorem 4.7 (Totalization closure schema). *Any operator purporting to canonically totalize/complete partial structure to generate new standing either introduces illicit primitives or collapses to bookkeeping-equivalence. Hence no admissible totalization exists that generates standing. In short: “canonical completion” that purports to create standing is either illicit primitive injection or mere re-expression.*

5 Main Theorem Arc

We state the main results here and provide full proofs in appendices.

Theorem 5.1 (No numeric coupling collapse). *Admissibility constraints applied to the RG equation do not force collapse of a continuous dimensionless gauge coupling to a unique numeric value g_* .*

Theorem 5.2 (Parameter-illusion classification). *Raw chart-numeric values of dimensionless gauge couplings are skin-level parameter-illusion labels and cannot, as raw numbers, be standing-bearing invariants. Redescription-invariant observable or compatibility data associated with an RG trajectory may still be tensor-eligible.*

Theorem 5.3 (Operator exhaustion). *No admissible operator-level strategy can generate uniqueness of a tensor identity by selection, typicality, completion, model-existence certification, meta-authority, forcing/genericity, totalization, structural-emergence detour, or by finite/iterative/diagonal/foundation-swap interaction among those closed families.*

Theorem 5.4 (Structural constraint reduction). *A structural or dynamical constraint in the fixed gauge-sector comparison class is not an additional uniqueness generator. It either restricts admissible redescription-invariant tensor data, changes the tensor package/object-class, or reduces to a closed representative-selection, witness, completion, or scope-transport route.*

Theorem 5.5 (Universal overlap necessity for co-authority). *If R_1 and R_2 are admissible and genuinely co-scoped (Definition D.1), then $\text{Dom}(R_1) \cap \text{Dom}(R_2) \neq \emptyset$.*

Source cross-reference. Universal overlap necessity is the overlap-facing form of the AASC no-selector/no-second-classifier spine: a pair of regimes that truly claims same-object co-authority cannot remain disjoint without a routing rule, and a nontrivial routing rule is exactly the kind of same-scope selector closed by Section 2 and Appendix C.

Theorem 5.6 (Forced-coincidence uniqueness on overlap). *Under universal overlap and same-object-class assertion on the overlap domain, there is at most one admissible representative on overlap (up to fixed normalization).*

Corollary 5.7 (Overlap-level representational uniqueness). *On any nonempty overlap domain where two admissible regimes assert the same standing identity class, the admissible representative is unique up to admissible redescription and fixed normalization.*

Corollary 5.8 (Physical-content uniqueness on overlap). *On such an overlap, the admissible physical content is likewise unique at the fixed comparison-class level: any alleged physical difference is either a tensor-level difference that cancels the same-object hypothesis or a skin-level difference without standing.*

Corollary 5.9 (Coupling non-uniqueness). *Numeric coupling parameters do not inherit overlap-level uniqueness; they remain skin-level chart labels unless proven invariant under admissible redescription.*

Proof pointers. Theorems 5.1–5.2 are proved in Appendices A–B. Theorem 5.3 and Theorem 5.4 are proved in Appendix C. Theorem 5.5 is proved in Appendix D. Theorem 5.6 and Corollaries 5.7–5.9 are proved in Appendix E.

6 Non-Claims

This paper does not:

- derive a unique numeric coupling constant;
- deny that coupling coordinates are useful for calculations or that invariant observables can depend on RG trajectory data;
- claim that changing physical boundary data cannot change predictions;
- invoke a UV completion as input;
- use empirical input as a selector;
- assert ontological unification;
- introduce predictive generative machinery;
- add new primitives;
- derive the Standard Model gauge group, matter content, or anomaly-free hypercharge assignments from nothing; or
- claim uniqueness of gauge theory *across all* possible gauge sectors in nature.

7 Conclusion

Minimal evaluability forces the same-object comparison discipline used here (Theorem 3.5), and redescription invariance follows from that discipline rather than from a protective definition (Theorem 3.6). Numeric coupling collapse fails (Theorem 5.1). Raw chart-numeric coupling values are skin-level parameter-illusion labels unless additional redescription-invariant tensor data are fixed (Theorem 5.2). All generative strategies for manufacturing standing-bearing uniqueness are exhausted and closed, including structural-emergence and interaction-generated variants (Theorems 5.3–5.4, Section 4). For genuinely co-scoped regimes (co-authority on a common object-class),

overlap is necessary (Theorem 5.5), and divergence on overlap is inadmissible, yielding forced-coincidence uniqueness of the admissible representative and of the admissible physical content on overlap (Theorem 5.6, Corollaries 5.7–5.8). Uniqueness, when present, is a closure certificate on overlap rather than a numeric parameter derivation.

Illustrative SM-like instance (non-exhaustive, non-derivational). As a familiar illustration of the overlap logic, suppose a chiral gauge tensor package is already fixed, including its representation content, anomaly-free compatibility conditions, locality/observable-compatibility constraints, and normalization conventions. Then any genuinely co-scoped admissible representation of that same package must coincide on the overlap up to admissible redescription and fixed normalization. This is not a derivation of the Standard Model from the closure rules, and it is not a derivation of the values of its gauge couplings. It is a conditional claim: once the tensor package and same-object-class assertion are fixed, overlap divergence is closed; the corresponding raw numeric coupling coordinates remain chart-level labels unless replaced by redescription-invariant observable data.

A Appendix A: Full RG Collapse Attempt

A.1 Renormalizable Gauge Theory Setup

Consider a renormalizable gauge theory with a single dimensionless coupling g , governed by the renormalization group equation

$$\mu \frac{dg}{d\mu} = \beta(g).$$

We analyze the ultraviolet limit $\mu \rightarrow \infty$.

The question examined is:

Does admissibility force collapse of continuous deformation of g to a unique numeric value g_* ?

A.2 Definition: UV Admissibility

A trajectory $g(\mu)$ is UV-admissible if and only if:

- (A1) The trajectory remains representable for all $\mu \geq \mu_0$ (no finite-scale obstruction).
- (A2) No standing violation occurs under infinite scaling.
- (A3) No illicit scope transport is required.
- (A4) Redescription of g does not alter tensor-level identity (standing identity is not a coordinate artifact).

Admissibility is binary; violations are terminal.

A.3 Finite-Scale Obstruction

Lemma A.1. *If the RG trajectory develops a finite-scale obstruction (e.g., terminates at some $\mu_L < \infty$), then the trajectory is UV-inadmissible.*

Proof. If $g(\mu)$ cannot be continued past μ_L , then the UV limit is undefined within the stated scope. Any attempt to appeal to an unspecified UV completion would introduce external structure beyond the scope (illicit scope transport). Hence admissibility fails. $\square$

A.4 Fixed Point Linearization

Suppose a UV fixed point exists:

$$\beta(g_*) = 0.$$

Linearize near g_* :

$$\mu \frac{d}{d\mu}(g - g_*) = \beta'(g_*)(g - g_*).$$

Solving yields:

$$g(\mu) - g_* = (g(\mu_0) - g_*) \left(\frac{\mu}{\mu_0} \right)^{\beta'(g_*)}.$$

A.5 Perturbation Behavior

Three cases arise:

Case 1: $\beta'(g_*) < 0$ (**UV attractive**). Then for sufficiently small perturbations $\delta g(\mu_0)$,

$$\delta g(\mu) \rightarrow 0 \quad \text{as } \mu \rightarrow \infty.$$

Thus a continuous family of initial conditions flows to the same fixed point; continuous deformation does not violate UV admissibility.

Case 2: $\beta'(g_*) > 0$ (**UV repulsive**). Only a tuned subset approaches g_* in the UV. But admissibility does not supply a rule that forbids the existence of multiple admissible trajectories; it only forbids violations (e.g., finite-scale obstruction, standing failure, scope transport). Therefore repulsiveness does not imply uniqueness.

Case 3: $\beta'(g_*) = 0$ (**marginal**). Marginal directions persist. RG structure alone does not eliminate them, and admissibility has not introduced an independent rule forbidding marginality *as such*. Hence no collapse follows.

A.6 Scheme dependence and invariants

One possible reading attempts to promote the numeric value g_* to a standing-bearing invariant. This is blocked by two standard facts:

- (A6.1) **Scheme dependence:** the beta function $\beta(g)$ and fixed-point location g_* generally depend on the renormalization scheme (reparameterization of couplings and subtraction conventions).
- (A6.2) **Universal data:** only scheme-invariant content (e.g., critical exponents such as $\beta'(g_*)$ in appropriately defined universality classes, or other properly invariant quantities) is eligible to be standing-bearing under Definition 3.3.

Thus the raw number g_* is not automatically eligible for standing.

A.7 Dimensional transmutation does not yield numeric collapse

A standard counterpressure is that running couplings can be exchanged for a scale via dimensional transmutation (e.g., in asymptotically free theories). A representative schematic relation is:

$$\Lambda \sim \mu \exp\left(-\frac{1}{b g^2(\mu)}\right),$$

for an appropriate constant b in the leading beta-function coefficient. This does *not* produce a unique numeric collapse of a dimensionless coupling; it trades a dimensionless choice for a dimensionful scale choice. It therefore does not contradict Theorem 5.1; it reinforces that uniqueness, if present, must be stated at the correct role level (tensor invariants), not as a bare number.

A.8 Redescription Invariance

Consider an admissible reparameterization (chart change)

$$\tilde{g} = f(g),$$

with f invertible on the admissible domain. Then

$$\tilde{\beta}(\tilde{g}) = \mu \frac{d\tilde{g}}{d\mu} = f'(g)\beta(g).$$

A fixed point g_* maps to $\tilde{g}_* = f(g_*)$. Hence numeric fixed-point values are chart-dependent unless proven invariant. This blocks promoting “the number g_* ” to standing identity without further structure.

A.9 Worked Example: Scheme Change and Invariant Extraction

We exhibit explicitly how coupling reparameterization changes raw coordinate data while preserving the invariant content that is eligible for standing.

Step 1: Perturbative beta function. Consider a one-coupling beta function through septic order:

$$\beta(g) = b_1 g^3 + b_2 g^5 + b_3 g^7 + O(g^9).$$

Step 2: Scheme redefinition. Let

$$\tilde{g} = g + c g^3 + d g^5 + O(g^7),$$

with constants c, d . This is invertible near $g = 0$ and is a chart change when tensor-compatibility relations are preserved.

Step 3: Transformed beta function. Using

$$\tilde{\beta}(\tilde{g}) = \frac{d\tilde{g}}{dg} \beta(g),$$

and expressing the result in the $\tilde{g}$ coordinate gives

$$\tilde{\beta}(\tilde{g}) = b_1 \tilde{g}^3 + b_2 \tilde{g}^5 + \tilde{b}_3 \tilde{g}^7 + O(\tilde{g}^9),$$

where

$$\tilde{b}_3 = b_3 - 2cb_2 + (2d - 3c^2)b_1.$$

Thus the first two coefficients are unchanged in this common perturbative redefinition class, while higher coefficients shift. This correction matters: the non-standing status of a raw coordinate value does not require pretending that every perturbative coefficient is scheme-dependent.

Step 4: Fixed-point coordinate shift. If a fixed point exists at g_* in the original chart, the corresponding fixed point in the new chart is

$$\tilde{g}_* = f(g_*) = g_* + cg_*^3 + dg_*^5 + O(g_*^7).$$

Unless f is numerically trivial at g_* , the raw number assigned to the fixed point changes. What is preserved is the standing identity of the fixed point under admissible redescription, not the coordinate label used to denote it.

Step 5: Invariant extraction and standing eligibility. Properly defined universal data, such as invariant critical exponents or invariant observable relations when their hypotheses are fixed, may be tensor-eligible under Definition 3.3. The raw coordinate g_* is not eligible merely because a chart writes it down.

Thus:

- the numeric coordinate g_* is a chart-dependent label;
- scheme-invariant data, when actually invariant under the admissible redescription class, are the eligible standing-bearing candidates.

A.10 Overlap Example: Coupling vs Transmutation Scale

We exhibit explicitly how two distinct parameterizations of the same theory coincide on overlap, illustrating forced-coincidence uniqueness without numeric collapse.

Step 1: Two parameterizations. Consider an asymptotically free theory with one-loop beta function

$$\beta(g) = -bg^3 + O(g^5), \quad b > 0.$$

Integration yields

$$\frac{1}{g^2(\mu)} = 2b \ln\left(\frac{\mu}{\Lambda}\right),$$

where Λ is the transmutation scale. Thus the theory may be parameterized equivalently by:

- an initial coupling value $g(\mu_0)$ at reference scale μ_0 ;
- the transmutation scale Λ .

These are related by

$$\Lambda = \mu_0 \exp\left(-\frac{1}{2bg^2(\mu_0)}\right).$$

Step 2: Overlap domain. The overlap domain consists of renormalization-group invariant observables (e.g. dimensionless ratios of physical quantities, or S-matrix elements evaluated at physical momenta). Such quantities are functions of either $g(\mu_0)$ or Λ , but not independently of both.

Step 3: Forced coincidence on overlap. Let R_1 be the regime using $g(\mu_0)$ as chart coordinate, and R_2 the regime using Λ as chart coordinate. On the overlap domain of physical observables, both regimes predict identical values:

$$\mathcal{O}(g(\mu_0)) = \mathcal{O}(\Lambda).$$

Any divergence would violate standing conservation (same claimed theory), redescription stability, and bivalence. Thus representational identity on overlap is unique up to normalization.

Step 4: Role classification. Under the role discipline:

- $g(\mu_0)$ is a chart coordinate on the RG trajectory;
- Λ is an alternative coordinate on the same trajectory once a scale convention is fixed;
- RG-invariant observable predictions and dimensionless physical ratios constitute tensor-level compatibility data.

A different value of the invariant trajectory data may describe a different tensor-level family member; the point here is that no raw chart number is thereby promoted into a unique standing certifier.

Conclusion. Continuous parameter freedom does not produce standing multiplicity merely by being written in different coordinates. Distinct chart parameterizations of the same trajectory coincide uniquely on overlap. Numeric collapse is unnecessary for representational uniqueness.

A.11 Non-perturbative effects do not force numeric coupling collapse

A possible objection is that non-perturbative physics (instantons, sphalerons, θ -vacua, topological sectors) might force a discrete “jump” that effectively collapses a continuous gauge coupling to a unique value. This does not occur without importing a selector.

Exhaustive classification. Non-perturbative structure can do one of the following:

- (NP1) **Create new observables/invariants:** e.g. topological sectors or global anomaly constraints. These are tensor-eligible candidates for standing-bearing quantities, but they are not numeric collapse of a continuous coupling.
- (NP2) **Create non-analytic dependence on existing parameters:** e.g. amplitudes with schematic dependence $\exp(-8\pi^2/g^2)$. Steepness or non-analyticity does not select a unique g absent boundary data.
- (NP3) **Introduce discrete superselection data:** e.g. θ as an independent coordinate. This changes the invariant structure rather than collapsing the original continuous coupling.

Selector analysis. A discrete admissibility set for g requires a rule of the form “only these values are admissible.” Such a rule can arise only from: (i) obstructions/singularities, (ii) quantization/topological classification, (iii) empirical selection, or (iv) an external selection principle. Obstructions exclude regions but do not pick a unique remaining point unless a singleton is already forced by tensor invariants; quantization changes invariant type rather than collapsing a chart parameter; empirical selection is out of scope; and external selection is closed by operator exhaustion.

Conclusion. Non-perturbative physics can contribute additional tensor-eligible invariants or modify functional dependence of observables on chart parameters, but it cannot generate numeric coupling collapse without importing a selector or changing the question type.

A.12 RG uniqueness routes are closed

Lemma A.2 (RG numeric-selector normal form). *Any RG-based argument that purports to output a unique raw chart-numeric coupling value must use at least one of the following routes:*

- (RG1) *a fixed-point coordinate read as a standing invariant;*
- (RG2) *boundary, initial-condition, matching, or empirical data selecting a trajectory;*
- (RG3) *an obstruction or quantization condition that removes all but one candidate;*
- (RG4) *a canonical scheme, normalization, metric, measure, or representative rule;*
- (RG5) *a non-perturbative or topological datum that changes the invariant structure;*
- (RG6) *a UV completion, model witness, or external authority that supplies the missing selection.*

Each route is either bookkeeping because the singleton was already fixed by tensor-invariant data and normalization, or inadmissible as a raw numeric-collapse argument because it imports a selector, changes scope, or mistakes a chart coordinate for standing content.

Proof. An RG derivation can affect a coupling coordinate in only three ways: it can evolve the coordinate along a trajectory, restrict the admissible set of trajectories, or redescribe the coordinate used to label the same trajectory. Evolution alone gives a flow, not a selector. Restriction to a singleton requires either an invariant obstruction/quantization condition, boundary data, matching data, empirical input, or an external certifier; those are exactly (RG2), (RG3), (RG5), and (RG6). Redescription supplies (RG1) and (RG4): a fixed-point value or canonical scheme can be treated as a raw number only after choosing a representative of the chart. If that representative is fixed by invariant tensor data and normalization, no new numeric-selection work has been done. If it is not so fixed, the representative rule is a selector, weighting, canonicalization, or imported witness, all closed by Theorem C.4. Thus every RG uniqueness route either changes the invariant problem or depends on a closed kernel. $\square$

Theorem A.3 (No admissible fundamental raw numeric coupling necessity). *Within the fixed gauge-sector comparison class, no interpretation can make a raw chart-numeric gauge-coupling value both fundamental-as-standing and uniquely forced unless that value is invariant under all admissible redescrptions or already fixed by tensor invariants plus normalization. In the first case it is no longer a raw chart number; in the second case the uniqueness is tensor-fixed bookkeeping. In every other case the interpretation violates redescription invariance or imports a selector.*

Proof. Assume a raw number g_C is declared fundamental and uniquely forced. If admissible scheme or chart change sends g_C to $\tilde{g}_C \neq g_C$ while preserving tensor compatibility, then the same standing identity receives two incompatible numeric standing verdicts. By Theorem 3.6, this is inadmissible unless a privileged chart is selected. A privileged chart is bookkeeping only when already fixed by tensor invariants and normalization; otherwise it is a representative-choice operator closed by Theorem C.4. If the purported numeric value is invariant under every admissible redescription, the standing-bearing object is the invariant content, not the raw chart label. Thus taking a coupling value as empirical input or boundary data is allowed, but elevating the raw number itself to necessary standing either changes the invariant problem or invokes a closed selector route. $\square$

A.13 No-Collapse Theorem

Theorem A.4 (No numeric collapse from RG+admissibility). *Admissibility constraints applied to the RG equation do not force collapse of a continuous dimensionless coupling to a unique numeric value.*

Proof. In the UV-attractive case, a continuous family of initial conditions is admissible and flows to the same UV fixed point, contradicting any claim that *any* deformation violates admissibility. In the marginal case, admissibility provides no mechanism eliminating marginality. In all cases, numeric coupling values are not guaranteed redescription-invariant (Sections A.6 and A.8), hence cannot be treated as standing-bearing by admissibility alone. Therefore no unique numeric value is forced. $\square$

A.14 Structural Diagnosis

The failure is structural:

- RG governs trajectories, not standing identity.
- Multiple admissible trajectories may share UV behavior.
- Numeric couplings are chart- and scheme-dependent absent further invariant proof.
- Dimensional transmutation trades dimensionless freedom for scale freedom; it does not certify a unique numeric coupling.
- Taking a coupling as input is compatible with ordinary gauge theory; what is closed is promotion of the raw chart number to a uniquely forced standing-bearing necessity.

Therefore numeric collapse is not the correct uniqueness target. This proves Theorem 5.1. The stronger interpretive restriction is Theorem A.3.

B Appendix B: Parameter-Illusion Theorem (Full Proof)

B.1 Standing-Bearing Criterion

We restate Definition 3.3: a quantity is standing-bearing only if it is invariant under admissible redescription and identity-determining at the tensor level.

B.2 Admissible Redescription (Role Preservation)

Admissible redescription is given by Definition 3.1: anchors fixed, tensor compatibility preserved, change confined to skin.

B.3 Chart-numeric coupling values

Let $g_{\mathcal{C}}(\mu_0)$ be the raw number assigned to a dimensionless coupling by a chosen chart/scheme $\mathcal{C}$ at a reference scale μ_0 . Under an admissible reparameterization,

$$\tilde{g} = f(g),$$

with f invertible on the admissible domain, the coordinate value changes while the standing identity of the represented RG trajectory is preserved whenever the transformation is a lawful chart change.

This appendix classifies the raw number $g_{\mathcal{C}}(\mu_0)$, not every invariant physical quantity that may be computed from an RG trajectory.

B.4 Non-Invariance of Raw Numeric Couplings

Lemma B.1. *Raw chart-numeric coupling values are not invariant under admissible redescription.*

Proof. Under $\tilde{g} = f(g)$ with $f \neq \text{id}$, the numerical label assigned to the same represented trajectory changes. If tensor compatibility and standing identity are preserved, the change is a lawful redescription. A standing-bearing quantity must be invariant under admissible redescription (Definition 3.3). Therefore the raw number g_C is not standing-bearing merely as a number. $\square$

B.5 Role-Diagnostic Strengthening

Lemma B.2. *Treating a redescription-variant coupling number as standing-bearing constitutes a role confusion ($\text{Skin} \rightarrow \text{Tensor}$) and is inadmissible under role discipline.*

Proof. If a quantity varies under admissible redescription while tensor compatibility is unchanged, then assigning it tensor-level standing forces a skin coordinate to carry tensor identity. This violates role preservation in Definition 3.1. Hence the move is inadmissible. $\square$

B.6 Continuous proximity cannot certify standing

Lemma B.3. *Any claim of “arbitrarily small” deformation δg presupposes a neighborhood notion for comparing parameter values.*

Proof. The phrase “arbitrarily small” is meaningful only relative to a notion of neighborhoods or limiting comparison. This is a proximity structure for the parameter labels. $\square$

Lemma B.4. *Proximity (“smallness”) of skin-level parameter labels cannot, by itself, certify standing identity.*

Proof. Standing identity is certified by invariance under admissible redescription and by tensor-compatibility effects (Definition 3.3). A proximity notion on skin labels is not, by itself, a tensor-compatibility invariant and can be altered by admissible reparameterization $\tilde{g} = f(g)$ (Section B.3). Therefore proximity of label values cannot be elevated to a standing certifier without importing extra structure beyond admissible invariants. $\square$

B.7 Parameter-Illusion Theorem

Definition B.5 (Chart-parameter illusion). *A raw parameter label is a chart-parameter illusion when its representative value can change under admissible redescription while the standing identity and tensor compatibility of the represented object remain fixed.*

Theorem B.6 (Parameter-illusion classification). *Raw chart-numeric values of dimensionless gauge couplings are chart-parameter illusions. They cannot be standing-bearing invariants as raw numbers.*

Proof. By Lemma B.1, raw coupling numbers vary under admissible redescription when the scheme/chart changes. By Lemma B.2, treating that redescription-variant number as tensor identity is a role confusion. By Lemma B.4, continuous proximity of such numbers supplies no standing certificate. Hence raw chart-numeric coupling values are chart-parameter illusions in the sense of Definition B.5. $\square$

B.8 Consequence

The question “what unique raw number must the coupling take” is ill-typed unless a redescription-invariant tensor datum and fixed normalization have already selected that number. If changing physical boundary data changes invariant observables, then the changed invariant data may be tensor-relevant; this does not rescue the raw coordinate as a standing certifier. Uniqueness, if present, must be formulated at the tensor-invariant level. This proves Theorem 5.2.

C Appendix C: Operator Exhaustion (Full Structural Elimination)

C.1 Exhaustion Framework

We examine whether standing-bearing uniqueness can be *generated* by any admissible operator-level maneuver or by a claimed structural/dynamical emergence route. The candidate families considered are: meta-authority re-anchoring, model-existence certification, compactness/saturation completion, typicality/measure selection, choice/selection, forcing/genericity, canonical totalization, and structural constraints alleged to create uniqueness without acting as an operator. These are the standard direct generative “escape hatches” by which one attempts to manufacture a distinguished object without a prior admissible invariant.

The AASC closure-by-exhaustion source adds the missing completeness burden: any non-bookkeeping same-scope operator exhibits an audited kernel, and every audited kernel is either one of the direct families or an interaction-generated variant of them. Finite composition, iteration, diagonalization, fixed-point detours, representation change, foundation swap, absoluteness, and transfer are therefore not additional escape hatches; they are routes back into the same closed registry. We show each family is inadmissible as a standing certifier, proving Theorem 5.3.

C.2 Registry completeness

Lemma C.1 (Audited-kernel completeness). *Every non-bookkeeping same-scope operator purporting to generate standing-bearing uniqueness depends on at least one audited kernel: a selector, witness, completion boundary, total evaluator, model/certifying structure, weighting rule, canonical representative rule, generic object, transfer criterion, repair rule, or meta-authority.*

Proof. If the operator is not bookkeeping, some datum must do standing-bearing work not already fixed by the input. If that datum were already fixed by admissible invariants, the operator would merely report fixed structure. Otherwise the datum is imported. The listed kernels exhaust the possible imported data used by a same-scope operator to turn non-unique input into a distinguished output. Interaction forms do not add a new datum type: the first non-bookkeeping stage in any pipeline already exhibits one of those kernels. $\square$

Corollary C.2 (No omitted operator family). *After the direct families and their interaction-generated variants are closed, no additional same-scope standing-generative operator family remains.*

Proof. By Lemma C.1, any omitted non-bookkeeping family would require an audited kernel. If the kernel is direct, it is treated in Sections C.3–C.9; if it appears only through interaction, it is still the first non-bookkeeping stage of a pipeline and reduces to a direct kernel. Therefore omission would require a standing-generative datum that is neither fixed nor imported, which is impossible. $\square$

Definition C.3 (Same-scope uniqueness generator). *A same-scope uniqueness generator is an operator or procedure that, without changing the fixed comparison class, purports to turn a non-singleton admissible domain of candidates into a unique standing-bearing representative.*

Theorem C.4 (Uniqueness-generator normal form). *Let F be a same-scope uniqueness generator on a non-singleton admissible domain D . Then F falls into exactly one of the following dispositions:*

- (U1) **Invariant-fixed singleton:** *admissible invariants of D already identify the unique representative, so F is bookkeeping.*
- (U2) **Invariant exclusion:** *tensor-level obstructions already remove all but one candidate; the uniqueness is a property of the fixed tensor package, not a generated representative.*
- (U3) **Representative choice:** *F selects, ranks, weights, measures, metrizes, canonicalizes, or privileges one candidate among admissibly undistinguished alternatives.*
- (U4) **Completion/witness import:** *F obtains uniqueness from a completion, total evaluator, compactness/saturation witness, model, generic object, UV completion, transfer principle, or meta-authority.*
- (U5) **Scope or target change:** *F changes the domain, carrier, tensor package, normalization, object-class, or standing quotient.*

Cases (U1)–(U2) do not generate standing-bearing uniqueness. Cases (U3)–(U4) are closed audited kernels. Case (U5) is not a same-scope counterexample.

Proof. A purported unique output can become unique only by changing one of three facts about D : which candidates are equivalent, which candidates are admissible, or which admissible representative is privileged. If equivalence or admissibility is changed by invariants already fixed in D , then the result is (U1) or (U2). If equivalence or admissibility is changed by material not fixed in D , then a new discriminator, boundary, witness, completion, or authority has been imported, giving (U3) or (U4). If no candidate is removed or identified but one is nevertheless returned as “the” representative, then the operation is direct representative choice, giving (U3). If the maneuver avoids all of these by altering the domain, carrier, object-class, normalization, or tensor package, then it has changed the target and falls under (U5). These alternatives are exhaustive because uniqueness of an output from a non-singleton domain is completely determined by admissibility, equivalence, and representative privilege. Therefore no fourth uniqueness-generating route remains. $\square$

Corollary C.5 (Completeness of the operator closure for this target). *For the fixed gauge-sector target, every same-scope uniqueness generator is either bookkeeping, tensor-fixed without generation, inadmissible by a closed audited kernel, or out of scope.*

Proof. Apply Theorem C.4. Cases (U1)–(U2) add no new standing-generative power. Cases (U3)–(U4) reduce to the closed families treated below. Case (U5) violates the fixed comparison class of Definition 2.1. $\square$

Theorem C.6 (Structural constraint reduction). *Let K be a structural or dynamical consistency condition offered as a same-scope source of uniqueness in the fixed gauge-sector comparison class. Then exactly one of the following holds:*

- (SC1) *K is expressible as a redescription-invariant restriction on tensor compatibility, in which case it performs invariant exclusion rather than generative selection;*
- (SC2) *K introduces new invariant structure, boundary data, object-class content, or normalization, in which case it changes the tensor package or target comparison class;*

- (SC3) *K privileges one representative among admissibly equivalent candidates, in which case it is a selector/canonicalization/measure route;*
- (SC4) *K obtains its force from a model, completion, UV completion, generic object, transfer principle, or external authority, in which case it is a witness/import route.*

Thus structural or dynamical emergence is not an omitted uniqueness-generator family.

Proof. A structural condition can affect the candidate domain only by changing admissibility, equivalence, or representative privilege. If it changes admissibility through already fixed tensor invariants, it is (SC1): the tensor package excludes candidates, and uniqueness, if any, was not generated by an added operator. If the condition requires a datum not already in the fixed package, that datum is new invariant structure, boundary data, or normalization, giving (SC2). If the condition leaves the invariant package unchanged but returns a preferred chart, scheme, representative, or numeric value, it has distinguished among admissibly equivalent representatives and is (SC3). If the condition is justified by an existence theorem, completion, UV completion, model witness, generic object, transfer criterion, or authority, it is (SC4). These alternatives exhaust the ways a constraint can alter a non-singleton candidate domain. Therefore any anomaly-cancellation, locality, RG, matching, UV, phase, or consistency condition is either a tensor restriction, a target change, or a closed audited kernel. $\square$

Corollary C.7 (No emergent-structure loophole). *Internal structural consistency may yield genuine tensor-level exclusion, including singleton exclusion, but it cannot turn a raw chart number into a standing-bearing unique value unless the relevant invariant data and normalization already fix that number.*

Proof. By Theorem C.6, a structural constraint either works at the invariant tensor level or leaves the fixed comparison class. A raw chart number varies under admissible redescription unless invariant data and normalization identify it. Therefore structural consistency can select tensor content but cannot by itself promote a redescription-variant coordinate into standing content. $\square$

C.3 Meta-Foundational Re-Anchoring

Definition C.8. *A meta-foundational re-anchoring operator is one whose action purports to certify admissibility/standing by appeal to a meta-authority external to its own standing-preserving action.*

Lemma C.9. *Any such operator imports standing not generated internally.*

Proof. The output's standing is mediated by the invoked authority. The operator does not generate that authority's standing. Hence standing is presupposed rather than produced. $\square$

Theorem C.10. *Meta-foundational re-anchoring cannot generate uniqueness as an admissible standing certifier.*

Proof. Immediate from Lemma C.9 and Theorem 4.1. $\square$

C.4 Model-Theoretic Existence

Definition C.11. *A model-theoretic existence operator is one whose action certifies existence/legitimacy by appeal to the existence of a model or certifying structure.*

Lemma C.12. *Such certification imports a witness whose standing is not generated by the operator.*

Proof. The existence claim is mediated by the model witness. The operator does not generate the standing of that witness; therefore standing is imported. $\square$

Theorem C.13. *Model-theoretic existence cannot generate uniqueness as a standing certifier.*

Proof. Immediate from Lemma C.12 and Theorem 4.2. $\square$

C.5 Compactness and Saturation

Definition C.14. *A compactness/saturation operator certifies existence/completeness by appeal to compactness or saturation principles as a generative component of its action.*

Theorem C.15. *Compactness or saturation cannot generate admissible uniqueness as a standing certifier.*

Proof. This is exactly Theorem 4.3. $\square$

C.6 Typicality and Measure

Definition C.16. *A typicality/measure operator certifies legitimacy/selection by appealing to probability, prevalence, typicality, or measure-theoretic weighting as a generative component of its action.*

Theorem C.17. *Typicality/measure cannot generate uniqueness as a standing certifier.*

Proof. This is exactly Theorem 4.4. $\square$

C.7 Choice and Selection Operators

Definition C.18. *A choice/selection operator selects a distinguished candidate from a non-singleton admissible domain D in the presence of multiple candidates not already uniquely fixed by admissible invariants of D .*

Lemma C.19 (Selection trichotomy). *Let S be any operator purporting to perform selection on an admissible domain D . Then at least one holds:*

- (T1) **Invariant-fixed (bookkeeping):** $S(D)$ is uniquely fixed by admissible invariants of D (hence bookkeeping-equivalent).
- (T2) **External-indexed (inadmissible):** $S(D)$ depends on an auxiliary selector/oracle/index not fixed by admissible structure of D .
- (T3) **Multiplicity distinction (inadmissible):** D exhibits nontrivial redescription multiplicity and S distinguishes among admissibly indistinguishable candidates.

Theorem C.20. *No admissible nontrivial selection operator exists; every such strategy is inadmissible or bookkeeping-equivalent.*

Proof. By Lemma C.19, the strategy falls into at least one case. Case (T1) is bookkeeping-equivalent (Definition 3.10). Cases (T2) and (T3) are inadmissible because they import external structure or distinguish indistinguishable candidates. This is exactly Theorem 4.5. $\square$

C.8 Genericity and Forcing

Definition C.21. *A genericity/forcing operator is any operator-level maneuver that purports to obtain generic power by outputting, asserting, stabilizing, or operationally exploiting a generic object.*

Theorem C.22. *Genericity/forcing cannot generate admissible uniqueness as a standing certifier.*

Proof. This is exactly Theorem 4.6. □

C.9 Operator Totalization

Definition C.23. *A totalization operator attempts to extend partial structure to a canonical total form that generates new standing.*

Theorem C.24. *No admissible canonical totalization exists that generates standing.*

Proof. This is exactly Theorem 4.7. □

C.10 Exhaustion Result

Theorem C.25 (Operator exhaustion). *All generative operator-level strategies for manufacturing standing-bearing uniqueness reduce to inadmissibility or bookkeeping-equivalence. Hence uniqueness of the admissible representative cannot be generated by any admissible operator-level strategy.*

Proof. By Lemma C.1 and Corollary C.2, any non-bookkeeping same-scope uniqueness generator must fall into the direct family registry, the structural-constraint reduction of Theorem C.6, or an interaction-generated variant of those routes. The direct families are treated in Sections C.3–C.9. Each is blocked as a standing certifier by the corresponding closure schema in Section 4, or collapses to bookkeeping-equivalence. Structural constraints either perform invariant exclusion, change the target, or reduce to those same kernels. Interaction variants inherit the first non-bookkeeping kernel of their pipeline and hence add no new admissible family. Therefore no admissible operator-level generator of standing-bearing uniqueness remains. □

This proves Theorem 5.3. Theorem 5.4 follows from Theorem C.6.

D Appendix D: Universal Overlap Derivation (Full Structural Proof)

D.1 Genuinely co-scoped regimes (co-authority)

Definition D.1 (Genuinely co-scoped regimes). *Two representational regimes R_1 and R_2 are genuinely co-scoped iff:*

- (D1) *both claim admissibility under the same anchor constraints and role discipline (Section 1, Definition 3.1), and*
- (D2) *both claim authority over at least one common interior object-class (i.e., there exists a would-be interior class for which each regime asserts standing-bearing applicability).*

Equivalently: R_1 and R_2 are not merely disjoint subtheories on disjoint domains; they assert co-authority over at least one common object-class within the same admissible interior.

Lemma D.2 (Co-scope burden). *The overlap theorem applies only when same-object co-authority is asserted. A pair of regimes that denies common object-class status exits the theorem’s target rather than refuting it.*

Proof. Definition D.1 is conditional on the regimes’ own co-authority claim. If the regimes do not assert authority over the same interior object-class, then no overlap conclusion is being demanded. If they do assert it, then the same-scope routing problem below is unavoidable. Thus co-scoping is not smuggled in as a conclusion; it is the hypothesis under which the theorem is activated. $\square$

Definition D.3 (Boundary-witnessed partition). *A proposed domain split between R_1 and R_2 is boundary-witnessed iff it supplies all of the following:*

- (W1) *an admissible invariant that fixes the applicability boundary without appeal to a further chooser;*
- (W2) *a proof that objects assigned to different sides are not the same standing object-class under the same interior authority, or else an admissible matching/translation domain on which same-object claims are compared;*
- (W3) *preservation of the fixed anchors, tensor package, standing quotient, and normalization conventions needed for the comparison.*

A boundary witness therefore either discharges the co-authority hypothesis by showing that the regimes are not genuinely co-scoped, or supplies the matching/translation overlap on which co-authority is evaluated.

Lemma D.4 (Assignment trichotomy for common object authority). *If two admissible regimes are presented as governing a common interior object-class while their domains are claimed to be disjoint, then exactly one of the following holds: the common-object assertion is withdrawn; a boundary-witnessed partition is supplied; or an unwitnessed assignment rule is required to decide which regime governs a would-be interior object.*

Proof. For an object in the alleged common class, applicability must either be denied as common, fixed by admissible boundary data, or decided without such data. Denial withdraws the co-authority hypothesis. Fixing by invariant boundary data is precisely a boundary-witnessed partition. Deciding without such data is an assignment rule from objects to regimes. The alternatives are exhaustive because every would-be object either remains in the common claim or does not, and if it remains, its regime applicability is either invariant-fixed or not. $\square$

Lemma D.5 (Contextual-validity evasion is closed). *Appealing to contextual validity, partial authority, scale separation, phase restriction, or domain-of-use language does not evade the overlap theorem. If the context boundary is fixed by admissible invariant data, it is a boundary witness and either removes genuine co-scoping or supplies a matching overlap. If it is not so fixed, it is hidden routing.*

Proof. A context label has admissibility-bearing force only if it decides applicability. If that decision is fixed by invariant boundary data satisfying Definition D.3, the co-scope hypothesis has been refined rather than contradicted. If the context label is not invariant-fixed, it chooses between R_1 and R_2 for the common class and is therefore a routing selector. Such routing is closed by Lemma D.8 below. $\square$

Lemma D.6 (Non-fragmentation of common object authority). *A regime cannot avoid overlap consequences by fragmenting its asserted authority over a common object-class into partial subdomains unless the fragmentation is boundary-witnessed. If two regimes still assert standing-bearing authority over the same object-class, then any unwitnessed partial-domain restriction is hidden routing rather than an admissible evasion.*

Proof. A partial-domain restriction has admissibility-bearing force only if it determines when the regime applies. If the restriction is fixed by invariant boundary data and either denies common-object status or supplies a matching/translation overlap, it is a boundary witness under Definition D.3; the overlap theorem is then either inapplicable because co-authority has been discharged, or it applies on the matching domain. If no such witness is supplied, the restriction must still decide, for a would-be common object, which regime governs. That decision is precisely the routing map of Section D.3. Since nontrivial routing is selection, unwitnessed fragmentation is inadmissible by Lemma D.8. Thus “partial theories” evade only by genuinely withdrawing same-object co-authority or by providing an admissible boundary witness; they do not refute overlap necessity. $\square$

D.2 The Routing Problem

Assume R_1 and R_2 are admissible and genuinely co-scoped. Suppose for contradiction:

$$\text{Dom}(R_1) \cap \text{Dom}(R_2) = \emptyset.$$

Then the purported common co-authority is represented by two disjoint domains, each governed by a different regime.

D.3 Necessity of a routing mechanism

To apply the co-authority claim (Definition D.1), one must determine, for each would-be interior object in the common object-class, which regime applies. This requires a routing rule

$$\Phi : (\text{interior objects in the common class}) \rightarrow \{R_1, R_2\}.$$

Lemma D.7. *Any such routing rule Φ constitutes a selection operator over an undistinguished multiplicity of candidate regimes.*

Proof. If R_1 and R_2 are both admissible and claim co-authority for the same interior class, then neither is distinguished by an admissible invariant of that class *prior* to the routing. The routing rule must choose between them for would-be interior objects. This is exactly selection from a non-singleton domain not uniquely fixed by admissible invariants, hence a choice/selection operator in the sense of Definition C.18. $\square$

D.4 Selection closure

Lemma D.8. *No admissible routing rule Φ exists that selects between disjoint genuinely co-scoped regimes.*

Proof. By Lemma D.7, Φ is a selection operator. By Theorem C.20, any such nontrivial selection operator is inadmissible or bookkeeping-equivalent. Routing that actually distinguishes between R_1 and R_2 is nontrivial; therefore it is inadmissible. $\square$

D.5 Evasion closures (formalized)

Lemma D.9 (Interface/transmissivity evasion is closed). *Attempting to replace routing by treating the boundary as a transmissive “interface” that filters objects into regimes is inadmissible.*

Proof. A transmissive interface introduces a mediating process or structure transport across the boundary. This mediation functions as a hidden routing operator, hence reintroduces selection under another name and violates Lemma D.8. $\square$

Lemma D.10 (Exterior-standpoint arbitration is closed). *Attempting to arbitrate regime applicability from an exterior standpoint is inadmissible.*

Proof. An exterior standpoint constitutes a vantage outside the admissible interior from which interior standing/routing is adjudicated. This is illicit scope transport: it presupposes the admissible interior while claiming to stand outside it to route interior objects. Hence inadmissible. $\square$

Lemma D.11 (Semantic repair is closed). *Attempting to resolve disjoint genuinely co-scoped regimes by semantic reclassification of scope is inadmissible.*

Proof. Scope is fixed by anchors and role discipline; admissible redescription (Definition 3.1) cannot alter role assignments or repair incompatibility by semantic maneuver. Therefore semantic reclassification is repair-by-redescription and inadmissible. $\square$

Lemma D.12 (Boundary-witness evasion is closed). *An alleged EFT/UV, chart/domain, phase/context, or regime-boundary split avoids co-scoping only by producing an admissible boundary witness in the sense of Definition D.3. Without such a witness, the split is hidden routing and is inadmissible.*

Proof. A witnessed partition changes the hypothesis: the regimes are not genuinely co-scoped on a common object-class without qualification. An unwitnessed partition still requires a rule deciding which regime governs an interior object. That rule is routing, hence selection, and is closed by Lemma D.8. $\square$

D.6 Universal overlap necessity

Theorem D.13 (Universal overlap necessity). *If R_1 and R_2 are admissible and genuinely co-scoped, then*

$$\text{Dom}(R_1) \cap \text{Dom}(R_2) \neq \emptyset.$$

Proof. Assume for contradiction that the domains are disjoint. By Lemma D.4, the common-object assertion must be withdrawn, a boundary-witnessed partition must be supplied, or a routing rule Φ is required. The first option cancels the theorem’s hypothesis rather than refuting it. The second either discharges genuine co-scoping or supplies a matching/translation overlap, again not giving disjoint co-authority. The third is impossible by Lemma D.8. Attempts to evade routing by interface mediation, exterior arbitration, semantic repair, contextual validity, boundary language, or partial-authority fragmentation are closed by Lemmas D.9–D.12 and Lemma D.6. Therefore disjoint genuinely co-scoped admissible regimes are impossible. Hence overlap is necessary. $\square$

This proves Theorem 5.5.

E Appendix E: Forced Coincidence Machinery (Full Closure Logic)

E.1 Overlap domain

Let R_1 and R_2 be admissible and genuinely co-scoped. By Appendix D, there exists a nonempty overlap domain

$$D = \text{Dom}(R_1) \cap \text{Dom}(R_2) \neq \emptyset.$$

E.2 “Same object-class on D ” (non-circular definition)

Definition E.1 (Same object-class on overlap). *We say R_1 and R_2 represent the same object-class on D iff for each $x \in D$ the representations $R_1(x)$ and $R_2(x)$ are asserted (by the regimes’ co-authority claims) to belong to the same standing identity class (Definition 3.2) under admissible redescription with fixed invariants.*

Boundary-witness criterion (EFT vs. UV completion). A common attempted evasion is to claim two regimes are “co-scoped in spirit” while treating them as disjoint by appeal to an EFT/UV split. In this paper, an EFT/UV pair avoids genuine co-scoping *only* by exhibiting an admissible *boundary witness*: an explicit domain partition (e.g. cutoff/validity boundary) together with an admissible matching/translation on the overlap of validity. Absent such a boundary witness, any denial of co-authority degenerates into an implicit routing rule (“which regime applies to a given interior object?”), which is a nontrivial selection operator and is closed by Lemma D.7 and Theorem C.20. Thus “EFT vs. UV” is not a loophole: it is either a formally witnessed scope partition (not co-scoped) or an illicit hidden routing (co-scoped and forced to overlap).

E.3 Admissible translation

An admissible translation $T : R_1 \rightarrow R_2$ must preserve standing identity classes and tensor compatibility (Definition 3.1); it introduces no new primitives.

E.4 Translation uniqueness lemma

Lemma E.2 (Uniqueness of admissible translation on overlap). *Let T and T' be admissible translations from R_1 to R_2 on the overlap domain D . Then for all $x \in D$,*

$$T(x) = T'(x)$$

up to fixed normalization.

Proof. Assume for contradiction that there exists $x \in D$ such that $T(x) \neq T'(x)$. By Definition E.1, both images purport to represent the same standing identity class. If they differ, an arbitration rule or privileged interpretation is required to decide which image is the correct representative. Either introduces illicit standing-bearing structure, violating the global constraints (Section 1). Therefore admissible divergence is impossible and $T = T'$ on D up to normalization. $\square$

E.5 Non-divergence principle

Lemma E.3 (Non-divergence under admissible composition). *No admissible representational regime admits divergent admissible representatives for the same standing identity class under admissible composition on an overlap domain.*

Proof. Assume divergence exists for the same standing identity class. Either both outcomes are treated as admissible (violating standing conservation) or one is privileged by an added rule (introducing a new primitive). Both violate the global constraints. Hence divergence is inadmissible. $\square$

E.6 Selection closure on overlap

Lemma E.4. *Selection operators cannot resolve divergence on overlap.*

Proof. Any attempt to resolve divergence by choosing one of two admissible candidates is a nontrivial selection operator on an undistinguished multiplicity. By Theorem C.20, such an operator is inadmissible or bookkeeping-equivalent. If bookkeeping-equivalent, it cannot distinguish the candidates; if nontrivial, it is inadmissible. Hence selection cannot resolve divergence. $\square$

E.7 Forced-coincidence uniqueness on overlap

Theorem E.5 (Forced-coincidence uniqueness on overlap). *Assume R_1 and R_2 are admissible and genuinely co-scoped, and assume they assert the same object-class on the overlap domain D (Definition E.1). Then there exists at most one admissible representative on D up to fixed normalization.*

Proof. Assume two distinct admissible representatives exist on D for the same standing identity class. Then there exist admissible translations T and T' that disagree on some $x \in D$, contradicting Lemma E.2. Alternatively, divergence violates Lemma E.3 and cannot be repaired by selection (Lemma E.4). Therefore at most one admissible representative remains on D up to normalization. $\square$

E.8 Physical-content bridge on overlap

Corollary E.6 (Physical-content uniqueness on overlap). *Under the hypotheses of Theorem E.5, the admissible physical content on D is unique at the fixed comparison-class level.*

Proof. Theorem E.5 gives a unique admissible representative up to admissible redescription and fixed normalization. By Theorem 3.9, same-object physical content is exactly the redescription-invariant tensor or observable content. Therefore two admissibly redescribed representatives cannot differ physically within the fixed comparison class. If an alleged difference is tensor-invariant, the same-object assertion of Definition E.1 has failed; if it is not tensor-invariant, it is skin and lacks standing. Hence admissible physical content is unique on the overlap. $\square$

E.9 Closure certificate classification

The uniqueness result in Theorem E.5:

- arises solely from admissibility and closure under co-authority on overlap,
- admits no alternative without standing violation,
- introduces no ontic or explanatory commitments,
- certifies representational coherence and, by Theorem 3.9, the corresponding admissible physical content on the fixed comparison class.

Hence it is classified as a closure certificate (forced coincidence on overlap).

This proves Theorem 5.6 and Corollaries 5.7–5.9.

F Appendix F: Exhaustion Checklist

This appendix records a compact checklist of standard operator-level “uniqueness generators,” together with their fail-closed disposition in this paper. The point is not to refute the underlying mathematics; it is to block these methods as *standing certifiers* under the global constraints in Section 1.

Proposed generator	Disposition
Nontrivial selection / choice	Inadmissible unless already fixed by admissible invariants; otherwise imports an oracle or distinguishes indistinguishables (Theorems 4.5, C.20).
Typicality / anthropic measure	Inadmissible as a standing certifier: weightings import standing not generated by admissible action (Theorem 4.4); cf. Weinberg (1987), Tegmark (2003) as representative measure-based proposals.
Compactness / saturation completion	Inadmissible as an existence/uniqueness certifier: completion imports standing beyond admissible action (Theorem 4.3).
Model-theoretic existence certification	Inadmissible as a standing certifier: “existence by model” imports a witness (Theorem 4.2).
Meta-foundational authority / re-anchoring	Inadmissible as a standing certifier: appeals to external authority import standing (Theorem 4.1).
Forcing / genericity	Inadmissible or bookkeeping-equivalent: generic output cannot generate standing-bearing operator power (Theorem 4.6).
Canonical totalization	Inadmissible or bookkeeping-equivalent: totalization that “creates” standing introduces illicit primitives (Theorem 4.7).

This table is intentionally minimal: the full structural eliminations are proved in Appendix C (and the overlap/forced-coincidence consequences in Appendices D–E). Representative totalization instincts appear across canonical-completion/maximality programs in foundations; the present claim is not about refuting those formalisms, but about blocking them as standing certifiers under the paper’s admissibility constraints.